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Anaerobic Antibiotics and the Risk of Graft-versus-Host Disease after Allogeneic Hematopoietic Stem Cell Transplantation

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ABSTRACT

Certain anaerobic bacteria are important for maintenance of gut barrier integrity and immune tolerance and may influence the risk of graft-versus-host disease (GVHD) after allogeneic hematopoietic stem cell transplantation (HSCT). We conducted a single-center retrospective cohort study of allogeneic HSCT recipients to evaluate associations between receipt of antibiotics with an anaerobic spectrum of activity and GVHD outcomes. We identified 1214 children and adults who developed febrile neutropenia between 7 days before and 28 days after HSCT and compared GVHD risk and mortality among patients who received anaerobic antibiotics (piperacillin-tazobactam or carbapenems; $n = 491$) to patients who received only antibiotics with minimal activity against anaerobes (aztreonam, cefepime, or ceftazidime; $n = 723$). We performed metagenomic sequencing of serial fecal samples from 36 pediatric patients to compare the effects of specific antibiotics on the gut metagenome. Receipt of anaerobic antibiotics was associated with higher hazards of acute gut/liver GVHD (hazard ratio [HR], 1.26; 95% confidence interval [CI], 1.03 to 1.54) and acute GVHD mortality (HR, 1.63; 95% CI, 1.08 to 2.46), but not chronic GVHD diagnosis (HR, 1.04; 95% CI: .84 to 1.28) or chronic GVHD mortality (HR, .88; 95% CI, .53 to 1.45). Anaerobic antibiotics resulted in decreased gut bacterial diversity, reduced abundances of Bifidobacteriales and Clostridiales, and loss of bacterial genes encoding butyrate biosynthesis enzymes from the gut metagenome. Acute gut/liver GVHD was preceded by a sharp decline in bacterial butyrate biosynthesis genes with antibiotic treatment. Our findings demonstrate that exposure to anaerobic antibiotics is associated with increased risks of acute gut/liver GVHD and acute GVHD mortality after allogeneic HSCT. Use of piperacillin-tazobactam or carbapenems should be reserved for febrile neutropenia cases in which anaerobic or multidrug-resistant infections are suspected.

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INTRODUCTION

Febrile neutropenia occurs in 85% to 100% of patients who undergo allogeneic hematopoietic stem cell transplantation (HSCT) [1,2]. Current guidelines from the Infectious Diseases Society of America and the National Comprehensive Cancer Network recommend that hospitalized patients with anticipated prolonged and severe neutropenia who develop fever

should receive an empirical broad-spectrum antibiotic with antipseudomonal activity, such as piperacillin-tazobactam, cefepime, ceftazidime, or a carbapenem (imipenem-cilastatin or meropenem) [3]. Although each of these antibiotics may affect the gut microbiota, only piperacillin-tazobactam and carbapenems have appreciable activity against anaerobic bacteria [4,5]. Recent studies suggest that receipt of these antibiotics during episodes of febrile neutropenia may increase the risk of graft-versus-host disease (GVHD) after allogeneic HSCT [6–8]. In addition, lower abundances of certain commensal gut anaerobes, particularly bacteria from the order Clostridiales,

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have been associated with the development of GVHD [8–10]. Many Clostridiales species produce butyrate, a short-chain fatty acid that is a preferred nutrient source for colonocytes and that is increasingly recognized to be important for maintenance of gut barrier integrity and immune tolerance [11–13]. Taken together, these data suggest that anaerobic antibiotics may increase the risk of GVHD after allogeneic HSCT through depletion of intestinal butyrate.

In this study, we identified 1214 children and adults who developed febrile neutropenia after allogeneic HSCT and evaluated associations between the receipt of broad-spectrum antibiotics and GVHD outcomes. Using shotgun metagenomic sequencing of serial fecal samples from 36 children and adolescents in this cohort, we determined the effect of these antibiotics on the diversity and composition of the gut microbiota and the relative abundance of bacterial genes for butyrate biosynthesis in the gut metagenome.

METHODS

Study Population

We performed this retrospective cohort study of patients undergoing their first allogeneic HSCT through the Duke Pediatric Blood and Marrow Transplant or Duke Hematologic Malignancies and Cellular Therapy Programs between January 1, 2005, and December 31, 2016. We excluded patients who received T cell-depleted grafts or who died within the first 28 days after HSCT. The study protocol was approved by the Duke University Institutional Review Board.

Transplantation Practices

Before February 2015, prophylaxis therapy with ciprofloxacin and metronidazole was routinely given to adult patients from the start of conditioning and continuing through the duration of neutropenia or initiation of i.v. broad-spectrum antibiotics. In February 2015, prophylaxis with ciprofloxacin alone became standard practice for adult patients. Routine antibacterial prophylaxis was not used for pediatric patients throughout the study period. For *Pneumocystis jirovecii* prophylaxis, pediatric and adult patients received trimethoprim-sulfamethoxazole from the start of conditioning and continuing until 2 days before HSCT, followed by trimethoprim-sulfamethoxazole, inhaled pentamidine, or i.v. pentamidine starting 30 days after HSCT. Initial therapy for grade II–IV acute GVHD and moderate to severe chronic GVHD was systemic corticosteroids throughout the study period. Oral budesonide was often added for patients with gastrointestinal involvement.

Data Sources and Measures

We extracted patient demographic information and HSCT characteristics from secure databases maintained by the transplantation programs. For acute GVHD, stage and grade were assigned using the criteria proposed by Przepiorka et al [14]. For chronic GVHD, stage and grade were assigned using the criteria established by the 2014 National Institutes of Health Consensus Development Project on Criteria for Clinical Trials in Chronic GVHD [15]. Transplantation providers recorded diagnoses of acute and chronic GVHD based on clinical symptoms and, when obtained, biopsy results. We considered GVHD to be the cause of death if it was classified by the transplantation physician as the primary cause of death or as a secondary cause of death in a patient who received treatment for GVHD within 30 days of death from an infection. We identified all antibiotics received by patients between 7 days before and 28 days after allogeneic HSCT using medication data extracted from electronic medical records.

Statistical Analyses

The analyses presented herein were limited to patients who received treatment for febrile neutropenia between 7 days before and 28 days after HSCT and were administered an antibiotic with an anaerobic spectrum of activity (ie, piperacillin-tazobactam or a carbapenem) or an antibiotic without appreciable anaerobic activity (ie, aztreonam, cefepime, or ceftazidime). Patients who were diagnosed with GVHD before the onset of febrile neutropenia were excluded. We used chi-square tests or 2-sample *t* tests to evaluate for differences in patient and HSCT characteristics in these groups. We used a Cox proportional hazards model to evaluate for an association between anaerobic antibiotics and our primary outcome of acute gut/liver GVHD. We similarly used Cox proportional hazards models to evaluate associations between receipt of anaerobic antibiotics and the following secondary outcomes: acute skin GVHD, acute GVHD mortality, chronic GVHD, and chronic GVHD mortality. We adjusted all models for the following potential confounding factors: year of HSCT, age, sex, race, indication for transplantation, HSCT donor, HSCT source, HLA matching, conditioning regimen intensity,

and GVHD prophylaxis regimen. The time at risk was defined as period from the date of HSCT until death from any cause or censoring. Patients were censored at 1 year after HSCT for acute GVHD outcomes and at 5 years after HSCT for chronic GVHD outcomes. The proportional hazards assumption was verified for each model using Schoenfeld residuals and an extended Cox model. For the models in which mortality was an outcome, we also performed a competing-risks analysis with a subdistribution hazard function [9]; however, the findings did not differ substantively from those obtained with Cox proportional hazards models. These analyses also showed that antibiotic group was not associated with mortality from causes other than GVHD.

Metagenomic Data Analyses

A subset of children and adolescents (<18 years) included in these analyses were enrolled in a prospective cohort study that began in October 2015. Subjects were recruited during the pretransplantation evaluation and were followed until 100 days after HSCT. For the present analyses, we identified 36 subjects who developed febrile neutropenia between 7 days before and 28 days after HSCT and received either an anaerobic antibiotic (piperacillin-tazobactam or meropenem) or cefepime. For each subject, we selected one fecal sample collected within 10 days before the initiation of antibiotics and a second fecal sample obtained after antibiotic therapy had been instituted for at least 48 hours. We extracted DNA using MagAttract PowerSoil DNA EP Kits (Qiagen, Germantown, MD). We then constructed DNA sequencing libraries using Nextera XT DNA Library Prep Kits (Illumina, San Diego, CA) and sequenced these libraries on a NextSeq500 Sequencing System as 150-bp end reads. We trimmed reads and removed host decontamination using Trimmomatic and Bowtie2 [16,17]. Host-decontaminated reads were profiled for bacterial species abundances using MetaPhlAn2 [18]. We aligned sequencing reads with a publicly available reference database to identify bacterial genes encoding butyrate biosynthesis enzymes [19]. We used Wilcoxon signed-rank tests to compare the Shannon diversity and Chao1 richness of paired fecal samples obtained before and during treatment with antibiotics. We compared the bacterial composition of paired fecal samples using PERMANOVA and Bray-Curtis distances. We used Wilcoxon signed-rank tests to compare the relative abundances of bacterial orders and genera in paired samples collected before and during antibiotic treatment. Trimmed and host-decontaminated metagenomic reads were then profiled for the abundance of genes involved in butyrate biosynthesis using ShortBRED [20]. Finally, we used Wilcoxon signed-rank tests to compare the abundances of genes involved in butyrate biosynthesis and detected by metagenomic sequencing in paired fecal samples across patient groups.

Data-Sharing Statement

The metagenomic sequencing dataset supporting the conclusions of this study is available in the Sequence Read Archive (PRJNA588783). The statistical files and script used for data analyses are also publicly available (https://github.com/mskelly7/Tanaka_GVHD_manuscript).

RESULTS

Patient Characteristics

Of 1214 patients with febrile neutropenia, 723 (60%) received aztreonam, cefepime, or ceftazidime only, and 491 (40%) received piperacillin-tazobactam or a carbapenem (Table 1). Compared with patients who received aztreonam, cefepime, or ceftazidime, patients who received an anaerobic antibiotic were older ($P < .0001$), more likely to have a hematologic malignancy ($P < .0001$), and more likely to have a peripheral blood donor source ($P < .0001$). The median duration of antibiotic therapy for febrile neutropenia in the aztreonam/cefepime/ceftazidime group was 17 days (interquartile range [IQR], 8 to 25 days). The median duration of antibiotic therapy for febrile neutropenia in the anaerobic antibiotic group was 15 days (IQR, 8 to 23 days), including 12 days (IQR, 7 to 18 days) on which an anaerobic antibiotic was received. Of 491 patients in the anaerobic antibiotic group, 293 (60%) received piperacillin-tazobactam and 275 (56%) received a carbapenem (Table 2).

GVHD Outcomes

Of the 1214 patients, 511 (42%) developed acute GVHD only, 88 (7%) developed chronic GVHD only, and 335 (28%) developed both acute and chronic GVHD. Overall, 270 of 491 patients (55%) who received an anaerobic antibiotic were diagnosed with grade II–IV acute GVHD, compared with 337 of 723

Table 1
Characteristics of the Study Population

Characteristic	Aztreonam, Cefepime, or Ceftazidime (N = 723)		Anaerobic Antibiotics (N = 491)		P Value
Age, yr, median (IQR)	14.4	(5.1–45.1)	37.4	(15.6–53.1)	<.0001
Sex, n (%)					.29
Female	287	(40)	210	(43)	
Male	436	(60)	281	(57)	
Race, n (%)					.66
Non-Hispanic white	538	(74)	356	(72)	
Black or African-American	108	(15)	90	(18)	
Latino or Hispanic-American	23	(3)	18	(4)	
Other	54	(7)	27	(5)	
Hematologic malignancy, n (%)	451	(62)	388	(79)	<.0001
Related HSCT donor, n (%)	243	(34)	139	(28)	.051
HSCT source, n (%)					<.0001
Peripheral blood	218	(30)	215	(44)	
Bone marrow	159	(22)	75	(15)	
Umbilical cord blood	346	(48)	201	(41)	
HLA matching, n (%)					.03
6/6	377	(52)	267	(54)	
5/6	155	(21)	76	(15)	
4/6 or lower	191	(27)	148	(30)	
Conditioning regimen, n (%)					.59
Myeloablative	570	(79)	381	(77)	
Nonmyeloablative or reduced-intensity	153	(21)	110	(22)	
GVHD prophylaxis regimen, n (%)					.01
Calcineurin inhibitor + methotrexate	226	(31)	165	(34)	
Calcineurin inhibitor + MMF	344	(48)	202	(41)	
Receipt of alemtuzumab or ATG	46	(6)	55	(11)	
Other	107	(15)	69	(14)	

MMF indicates mycophenolate mofetil; ATG, antithymocyte globulin.

(46%) who received aztreonam, cefepime, or ceftazidime. Details on organ involvement and stages of acute GVHD diagnoses by antibiotic group are provided in Supplementary Table S1. Four hundred thirty-one patients (36%) were diagnosed with acute GVHD of the gut or liver (Table 3). Cumulative incidence curves for GVHD outcomes by antibiotic group are shown in Figure 1. In multivariable analyses, receipt of an anaerobic antibiotic regimen was associated with a higher hazard of acute GVHD of the gut or liver (hazard ratio [HR], 1.26; 95% confidence interval [CI], 1.03 to 1.54) and 1-year acute GVHD mortality (HR, 1.63; 95% CI, 1.08 to 2.46). Receipt of anaerobic antibiotics was not associated with acute skin GVHD (HR, 1.06; 95% CI, .86 to 1.31), chronic GVHD (HR, 1.04; 95% CI, .84 to 1.28), or 5-year chronic GVHD mortality (HR, .88; 95% CI, .53 to 1.43). The HRs for these analyses were similar when patients from the pediatric and adult HSCT programs were analyzed separately (Supplementary Table 2).

Table 2
Antibiotic Regimens for Febrile Neutropenia

Regimen	N	(%)
Aztreonam/cefepime/ceftazidime group	723	(60)
Received cefepime or ceftazidime only	684	(56)
Received aztreonam only	39	(3)
Anaerobic antibiotic group	491	(40)
Received piperacillin-tazobactam only	216	(18)
Received a carbapenem only	198	(16)
Received both piperacillin-tazobactam and a carbapenem	77	(6)

Effect of Antibiotics on the Diversity and Composition of the Gut Microbiota

Of the 36 children and adolescents included in this analysis, 26 received cefepime and 10 received piperacillin-tazobactam or meropenem (Supplementary Table S3). There were no culture-proven infections that corresponded with the timing of initiation of antibiotics in these 36 subjects. Fecal samples obtained following initiation of antibiotic therapy were collected after a median of 10 days (IQR, 6 to 27.5 days) among subjects receiving cefepime and a median of 14.5 days (IQR, 10 to 22 days) among subjects receiving piperacillin-tazobactam or meropenem (Wilcoxon rank-sum test; $P = .30$). No subject received clindamycin, metronidazole, or another antibiotic with substantial anaerobic activity before fecal sample collection. A total of 637,698,091 high-quality metagenomic sequences (mean of 8,856,918 sequences per sample) were obtained from the 72 fecal samples included in this analysis. Sequences were assigned to 284 bacterial species representing 116 genera from 27 orders and 6 phyla. Among subjects receiving anaerobic antibiotics, the median Shannon diversity declined from 2.0 (IQR, 1.6 to 2.5) before antibiotics to .9 (IQR, .3 to 1.2) while on antibiotic therapy ($P = .01$) (Supplementary Figure S1). Median Chao1 richness also declined from 28 (IQR, 19 to 49) to 5 (IQR, 3 to 15) with exposure to anaerobic antibiotics ($P = .01$). In comparison, cefepime treatment was not associated with changes in gut bacterial diversity or species richness. Bacterial composition differed in samples collected before and during anaerobic antibiotic therapy ($P < .001$) and before and during treatment with cefepime ($P = .01$). Cefepime and anaerobic antibiotics were associated with distinct changes in the

Table 3
GVHD Outcomes by Antibiotic Group

Outcome	Aztreonam, Cefepime, or Ceftazidime (N = 724), n (%)		Anaerobic Antibiotics (N = 492), n (%)		Adjusted HR (95% CI)		P Value
Primary outcome							
Acute gut/liver GVHD	227	(31)	204	(41)	1.26	(1.03–1.54)	.02
Secondary outcomes							
Acute skin GVHD	368	(51)	248	(51)	1.05	(.89–1.25)	.55
1-year acute GVHD mortality	45	(6)	55	(11)	1.63	(1.08–2.46)	.02
Chronic GVHD	264	(36)	159	(32)	1.04	(.84–1.28)	.73
5-year chronic GVHD mortality	47	(7)	25	(5)	.88	(.53–1.45)	.61

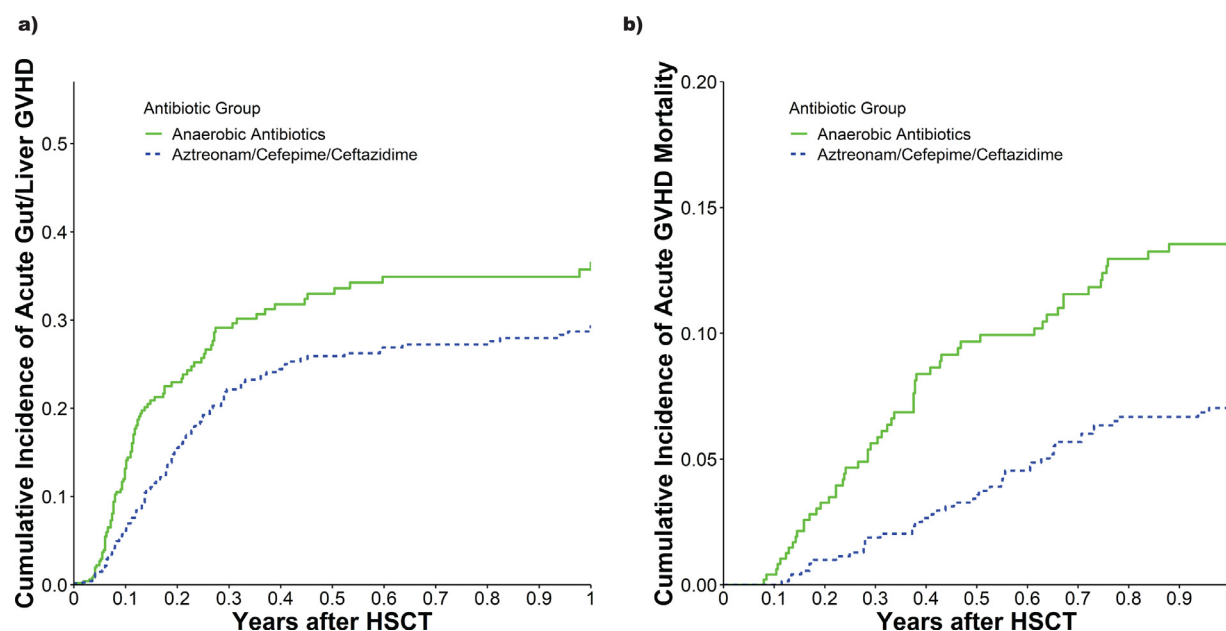


Figure 1. Cumulative incidence curves for GVHD outcomes by antibiotic group: (A) acute gut/liver GVHD; (B) acute GVHD mortality. In multivariable Cox proportional hazards models, the hazards of acute gut/liver GVHD ($P = .02$) and acute GVHD mortality ($P = 0.02$) differed by antibiotic group.

relative abundances of bacterial orders (Figure 2 and Supplementary Figure S2). Cefepime treatment was associated with declines in the relative abundances of Enterobacterales ($P = .01$) and Lactobacillales ($P = .04$). In contrast, anaerobic antibiotic therapy was associated with a decline in the relative abundances of Bifidobacteriales ($P = .01$) and Clostridiales ($P = .002$) and an increase in the relative abundance of Lactobacillales ($P = .01$). Several genera of Clostridiales declined in abundance during treatment with anaerobic antibiotics (Table 4), including genera that contain known butyrate-producing species (ie, *Blautia*, *Clostridium*, and *Lachnoclostridium*). In addition, *Bifidobacterium* was among the most abundant genera in fecal samples before antibiotic therapy but was not identified in any samples collected during anaerobic antibiotic treatment.

Genes for Butyrate Biosynthesis and Acute GVHD of the Gut or Liver

The relative abundance of Clostridiales in the gut microbiota declined in all patients who received piperacillin-tazobactam or meropenem, including patients who did not subsequently develop acute GVHD of the gut or liver (Figure 3). The abundance of genes for butyrate biosynthesis also declined among patients receiving anaerobic antibiotics ($P = .01$), from a median of 203 (IQR, 63 to 514) reads per kilobase million (RPKM) before antibiotics to 0 (IQR, 0 to 10) RPKM during

treatment with anaerobic antibiotics. In contrast, the median RPKM of butyrate biosynthesis genes did not differ between samples collected before and during treatment with cefepime (151 [IQR, 40 to 351] versus 62 [IQR, 27 to 209]; $P = .22$). Among patients receiving anaerobic antibiotics or cefepime, the abundance of genes for butyrate biosynthesis was strongly correlated with subsequent onset of acute gut/liver GVHD. Specifically, patients who later developed acute gut/liver GVHD had higher abundances of butyrate biosynthesis genes before antibiotic therapy (median RPKM, 512 [IQR, 319 to 799] versus 94 [IQR, 37 to 151]; $P < .0001$) and a more substantial decline in butyrate biosynthesis gene abundance with antibiotic treatment (median RPKM, 512 [IQR, 319 to 799] to 29 [IQR, 3 to 64]; $P = .0001$).

DISCUSSION

In this study of children and adults with febrile neutropenia after allogeneic HSCT, we found that receipt of piperacillin-tazobactam or a carbapenem increased the risk of acute GVHD of the gut or liver and acute GVHD mortality. In a small group of pediatric patients in this study, receipt of anaerobic antibiotics was associated with a loss of bacterial genes for butyrate biosynthesis from the gut metagenome, a finding that preceded the onset of acute gut/liver GVHD.

Our findings are consistent with several recent studies that suggest that use of antibiotics with an anaerobic spectrum of

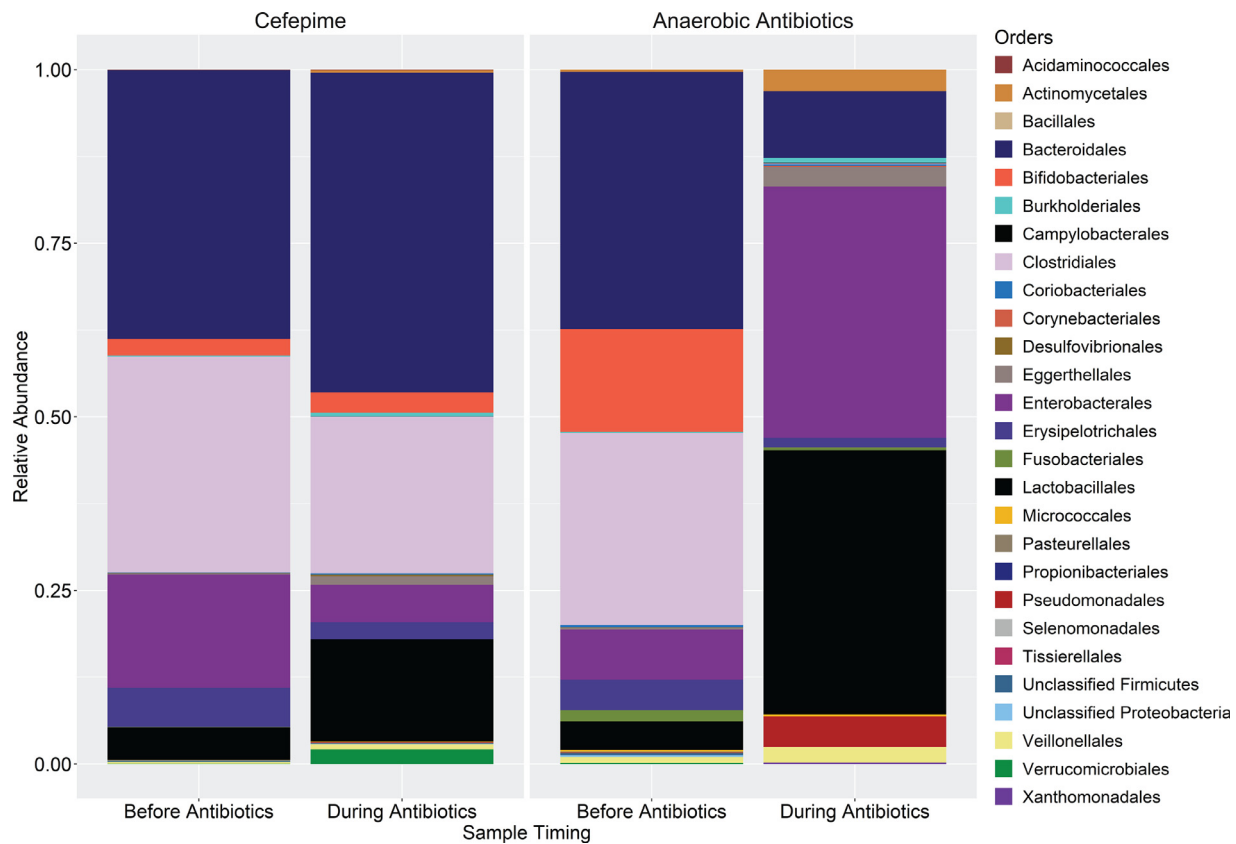


Figure 2. Relative abundances of bacterial orders in fecal samples collected from children before and during antibiotic treatment. Bar plots depict the mean relative abundances of bacterial orders in fecal samples collected from 36 children and adolescents within the 10 days before the initiation of antibiotics (before antibiotics) and while on antibiotic therapy for at least 48 hours (during antibiotics). The relative abundances of Bifidobacteriales ($P = .01$) and Clostridiales ($P = .002$) declined in children receiving anaerobic antibiotics.

activity increases GVHD risk after allogeneic HSCT. Shono et al [6] reported that in a series of 383 adult allogeneic HSCT recipients, empirical treatment of febrile neutropenia with piperacillin-tazobactam or imipenem-cilastatin was associated with an elevated risk of combined (acute and chronic) GVHD mortality compared with aztreonam or cefepime. Using a murine model of allogeneic HSCT, the authors demonstrated that treatment with piperacillin-tazobactam or imipenem-cilastatin was associated with loss of Clostridiales from the gut microbiota, whereas treatment with aztreonam or cefepime generally spared this bacterial order [6]. In 211 adults who underwent allogeneic HSCT, receipt of carbapenem antibiotics (compared with no antibiotics) during the neutropenic period was associated with the development of acute gut GVHD [7].

Finally, in a study of 15 pediatric allogeneic HSCT recipients at 2 institutions, the duration of anaerobic antibiotics treatment for febrile neutropenia was associated with acute GVHD risk [8]. Notably, these studies had small sample sizes, did not directly compare anaerobic antibiotics to other broad-spectrum antibiotic regimens, or did not account for key confounding factors, such as GVHD prophylaxis regimen. Our study overcomes these limitations in demonstrating that the use of anaerobic antibiotics for febrile neutropenia after allogeneic HSCT is specifically associated with the development of acute GVHD of the gut or liver. This finding was observed in both pediatric and adult allogeneic HSCT recipients and despite the differing use of antibiotic prophylaxis in these populations. In contrast, exposure to anaerobic antibiotics was not associated

Table 4
Effect of Anaerobic Antibiotics on the Relative Abundances of Frequently Occurring Bacterial Genera from the Orders Bifidobacteriales and Clostridiales

Class/Genus	Relative Abundance, %, median (IQR)		P Value
	Before Anaerobic Antibiotics	During Anaerobic Antibiotics	
Bifidobacteriales			
<i>Bifidobacterium</i>	9.7 (.06-24.3)	0 (0-0)	.02
Clostridiales			
<i>Blautia</i>	4.1 (.05-9.1)	0 (0-0)	.01
<i>Clostridium</i>	1.3 (.02-6.7)	0 (0-0)	.02
<i>Faecalibacterium</i>	0 (0-2.4)	0 (0-0)	.18
<i>Lachnoclostridium</i>	.2 (.03-1.5)	0 (0-0)	.01
<i>Roseburia</i>	.002 (0-2)	0 (0-0)	.06
<i>Ruthenibacterium</i>	.09 (0-.6)	0 (0-0)	.06

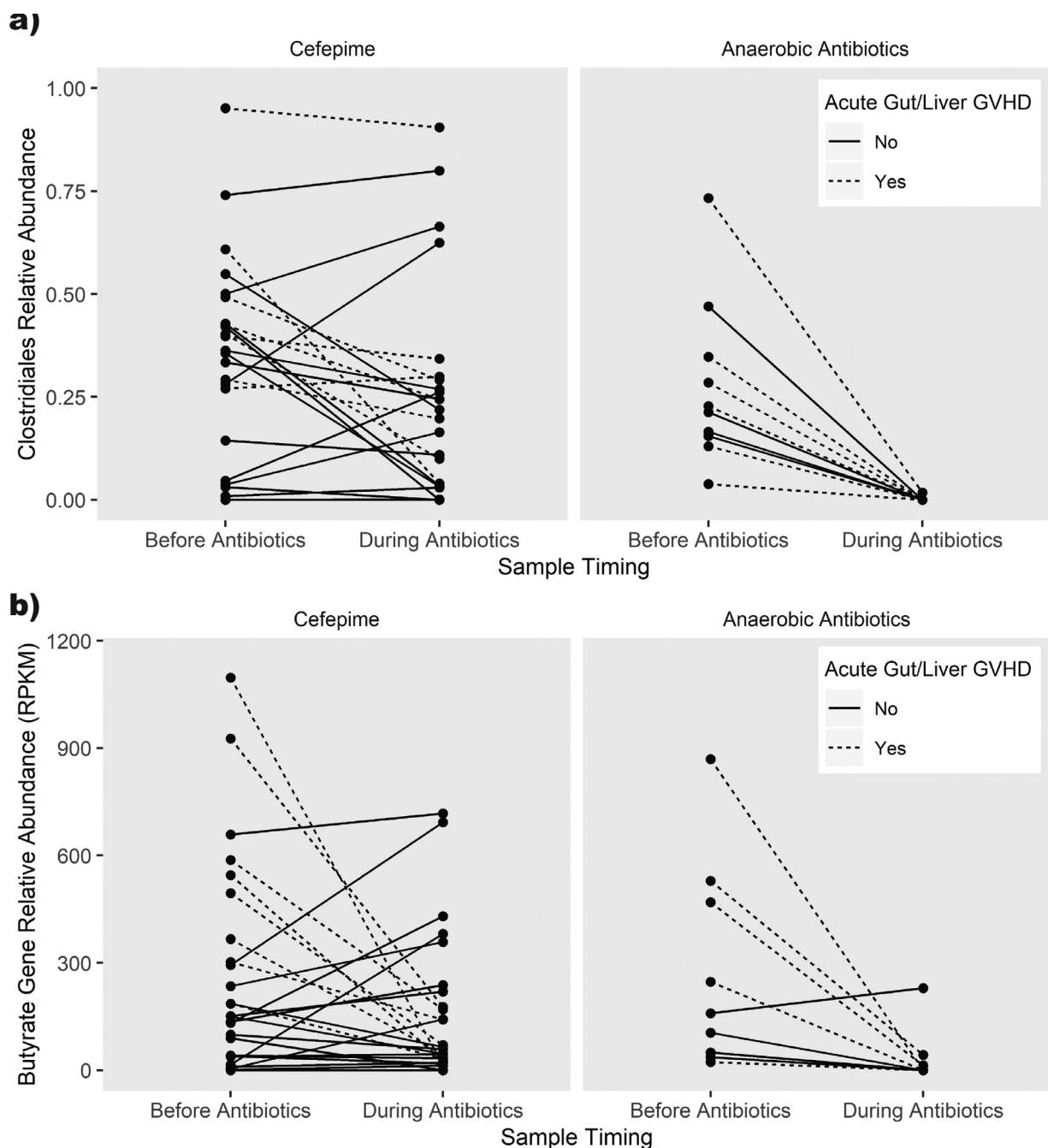


Figure 3. Changes in the abundances of Clostridiales and genes for butyrate biosynthesis during antibiotic treatment. Relative abundances of the bacterial order Clostridiales (A) and genes for butyrate biosynthesis (B), measured as RPKM, are shown for patients before treatment and during treatment with cefepime or anaerobic antibiotics. Each point represents the relative abundance of a single fecal sample, and lines are drawn between fecal samples collected within the 10 days before that initiation of antibiotics (before antibiotics) and fecal samples collected while on antibiotic therapy for at least 48 hours (during antibiotics) for each patient. Samples from patients who subsequently developed acute gut/liver GVHD are shown with a dashed line, and samples from patients who did not develop acute gut/liver GVHD are shown with a solid line. The relative abundance of Clostridiales in the gut microbiome ($P = .002$) and the abundance of genes for butyrate biosynthesis ($P = .01$) declined among children receiving anaerobic antibiotics. Butyrate biosynthesis gene abundance also declined markedly during antibiotic treatment in children who later developed acute gut/liver GVHD ($P = .0001$).

with chronic GVHD in our cohort. Although relatively little is known about the pathophysiology of chronic GVHD, B cells are believed to play a prominent role, and it is possible that the gut microbiome has less influence on the number and function of these cells [21].

Commensal gut anaerobes play an important role in maintaining intestinal homeostasis, and a number of recent studies have suggested that these bacteria may modify the risk of GVHD. In particular, a reduction in bacterial diversity and

losses of obligate anaerobic bacteria from the order Clostridiales have previously been associated with the risk of GVHD [8,9,22–24]. Clostridiales prevent inflammation and promote immune tolerance in the gut through the production of butyrate, which induces the differentiation of intestinal regulatory T cells [10,25,26]. Regulatory T cells are potent suppressors of immune responses and are believed to play an important role in mitigating the severity of GVHD after allogeneic HSCT [27,28]. Loss of intestinal butyrate after allogeneic HSCT was

previously demonstrated with exposure to anaerobic antibiotics and was associated with the risk of development of GVHD in a study of 42 pediatric HSCT recipients [29]. Moreover, butyrate was reduced in the intestinal tissue of mice after allogeneic HSCT, and the administration of butyrate to these mice via intragastric gavage lowered GVHD clinical scores and improved survival [10]. In contrast, intestinal butyrate levels at the time of conditioning and 2 weeks after HSCT were not associated with acute GVHD in a study of 50 adult allogeneic HSCT recipients [30]. Our findings are broadly consistent with these previous studies while indicating that the magnitude of the change in intestinal butyrate levels resulting from broad-spectrum antibiotics may be more important than the absolute level. In addition, in our cohort, anaerobic antibiotic treatment was associated with a reduction in the relative abundance of *Bifidobacterium* in the gut microbiota. Although bifidobacteria do not directly produce butyrate, their role in promoting the growth of Clostridiales has been described in detail [31,32].

Our results have several important implications for the care of patients undergoing allogeneic HSCT. First, we present data from the largest cohort to date demonstrating that the use of anaerobic antibiotics for febrile neutropenia increases acute GVHD risk and mortality in pediatric and adult allogeneic HSCT recipients. Our findings, when combined with previous data from other institutions, indicate that the use of piperacillin-tazobactam and carbapenems should be reserved for cases in which anaerobic or multidrug-resistant infection is suspected. In addition, this study demonstrates that measurement of the bacterial genes that encode butyrate biosynthesis enzymes through shotgun metagenomics may provide useful information for assessing patients' risk for acute GVHD of the gut or liver. Although such methods as high-performance liquid chromatography enable direct measurement of butyrate in stool, these approaches are labor-intensive, expensive, and unlikely to be feasible in routine clinical practice [33,34]. Finally, we found that patients who developed acute gut/liver GVHD had greater declines in the abundance of butyrate biosynthesis genes in the gut metagenome with exposure to broad-spectrum antibiotics. Dramatic shifts in the diversity and composition of the gut microbiota occur after allogeneic HSCT, and it is likely that serial monitoring of the gut metagenome and metabolome will be necessary to accurately forecast GVHD and other outcomes after allogeneic HSCT [35,36].

This study has several limitations. First, it was conducted at a single academic hospital, the transplantation practices of which may differ from those at other centers. Moreover, given the observational study design, patients were not randomly assigned to antibiotic treatments, and the potential for unmeasured confounding by indication exists. In addition, although GVHD diagnoses were made by the treating physicians using standardized criteria, this process remains somewhat subjective. We did not record medications administered for the treatment of GVHD, although initial therapy with glucocorticoids was standard throughout the study period at our institution. We reviewed autopsy reports, provider notes, and the results of microbiological testing for all deceased subjects, but misclassification of the cause of death remains possible. For consistency across the study period, we determined the degree of HLA matching based only on allele-level typing of HLA-A, -B, and -DRB1. Typing of HLA-C and -DQB1 is currently being performed at our institution and many other transplantation centers. Metagenomic analyses were limited to the small number of pediatric patients for whom paired fecal samples were available. Previous studies indicate that the composition of the gut microbiome of healthy children differs from that of healthy

adults [34]. Finally, we used detection of butyrate biosynthesis genes as a surrogate marker for the functional capacity of the microbiota to produce butyrate. There are few available data to suggest that these gene abundances correlate closely with actual butyrate concentrations in the gut. Moreover, recent studies suggest that other host or gut microbiome-derived metabolites may be associated with onset of acute or chronic GVHD after allogeneic HSCT, including bile acids and branched-chain amino acids [37,38].

In summary, we found that the use of anaerobic antibiotics for the empirical treatment of febrile neutropenia was associated with increased risks of acute GVHD of the gut or liver and acute GVHD mortality in adult and pediatric HSCT recipients. Moreover, we demonstrate that exposure to anaerobic antibiotics is associated with loss of genes for butyrate biosynthesis from the gut metagenome, which preceded the onset of acute gut/liver GVHD in our cohort. Our findings indicate that limiting the use of piperacillin-tazobactam and carbapenems to clinical scenarios in which anaerobic or multidrug-resistant infections are suspected could lower GVHD-related mortality in allogeneic HSCT recipients.

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SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:[10.1016/j.bbmt.2020.07.011](https://doi.org/10.1016/j.bbmt.2020.07.011).

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